

JUNGMIN KIM, PH.D.

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EDUCATION

Ph.D. Electrical and Computer Engineering

Seoul National University

Advisors: Prof. Namkyoo Park & Prof. Sunkyu Yu

Mar 2017–Feb 2023

Seoul, South Korea

B.S. Electrical and Computer Engineering

Seoul National University

Mar 2012–Feb 2017

Seoul, South Korea

WORK EXPERIENCE

University of Wisconsin-Madison

Research Associate, Scientist I

Madison, WI, USA

Apr 2023–Present

- Mentor: Prof. Zongfu Yu
- Designed mmWave face verification system using metasurface, with $> 90\%$ accuracy.
- Designed incoherent meta-imaging system for low-light environment, with ~ 2.9 dB enhancement of PSNR.
- Designed photonic systolic array for all-optical matrix-matrix multiplication, achieving theoretical computing density of $4.4 \text{ PMACS} \cdot \text{mm}^{-2} \text{s}^{-1}$.

Seoul National University

Research/Teaching Assistant

Seoul, South Korea

Mar 2017–Feb 2023

- Designed tilted Dirac cones and investigated temporal disordered modulation of photonic system to achieve target scattering.
- ECE courses on Quantum Mechanics, Electromagnetism, and Nanophotonics.

SKILLS

EM simulations	FDTD (Tidy3D), FEM (COMSOL), RCWA, PWEM
Programming	Matlab, Python
Optimization	Adjoint method, Automatic differentiation (Autograd), Gradient-based inverse design
Machine learning	PyTorch

SELECTED PROJECTS

Optical computing

Related publications: [1–4, 6]

Using gradient-based optimization techniques, including the adjoint method and neural network modeling of diffractive optical systems, I have inverse-designed various optical devices for neuromorphic and computing applications. These include neuromorphic metasurfaces for face recognition systems and output-stationary optical systolic arrays that accelerate matrix–matrix multiplication.

Disordered photonics

Related publications: [5, 7, 8]

Harnessing hidden degrees of freedom in material phase can unlock numerous functionalities. I used a deep learning approach to inverse-design photonic multilayer structures with angle-dependent optical target re-

ponses and active tunability. On the other hand, I extended the concept of correlated disorder—an intermediate regime between perfect order and complete disorder—into the temporal domain, enabling top-down control of light scattering through time-varying disordered modulation without the need for complex material structures.

Photonic band structure engineering

Related publications: [9, 10]

The photonic Dirac cone enables exceptional light propagation with intriguing applications such as optical cloaking and light funneling, while its directional control has remained largely unexplored. To address this, I developed an inverse design approach that perturbs the eigenmodes of a conventional upright Dirac cone to create a tilted version, enabling more efficient and directional light transport in photonic devices.

PUBLICATION

- ¹Q. Zhou, J. Kim, Y. Tao, et al., “All-optical deep learning with quantum nonlinearity”, [arXiv, 2601.01690 \(2026\)](#).
- ²J. Kim, N. Yu, and Z. Yu, “Compute-first optical detection for noise-resilient visual perception”, [ACS Photonics](#) **12**, 1137–1145 (2025).
- ³J. Kim, Q. Zhou, and Z. Yu, “Photonic systolic array for all-optical matrix-matrix multiplication”, [Laser & Photonics Reviews](#), e01995 (2025).
- ⁴J. Kim, J.-Y. Kim, J. Kim, et al., “Inverse design of nanophotonic devices enabled by optimization algorithms and deep learning: recent achievements and future prospects”, [Nanophotonics](#) **14**, 121–151 (2025).
- ⁵J. Kim, D. Lee, S. Yu, and N. Park, “Unidirectional scattering with spatial homogeneity using correlated photonic time disorder”, [Nature Physics](#) **19**, 726–732 (2023).
- ⁶S. Choi, J. Kim, J. Kwak, N. Park, and S. Yu, “Topologically protected all-optical memory”, [Advanced Electronic Materials](#) **8**, 2200579 (2022).
- ⁷J. Kim, S. Park, S. Yu, and N. Park, “Machine-engineered active disorder for digital photonics”, [Advanced Optical Materials](#) **10**, 2102642 (2022).
- ⁸S. Oh, J. Kim, X. Piao, et al., “Control of localization and optical properties with deep-subwavelength engineered disorder”, [Optics Express](#) **30**, 28301–28311 (2022).
- ⁹S. Park, I. Lee, J. Kim, N. Park, and S. Yu, “Hearing the shape of a drum for light: isospectrality in photonics”, [Nanophotonics](#) **11**, 2763–2778 (2022).
- ¹⁰J. Kim, S. Yu, and N. Park, “Universal design platform for an extended class of photonic dirac cones”, [Physical Review Applied](#) **13**, 044015 (2020).

PRESENTATION

- ^{C1}J. Kim, N. Yu, and Z. Yu, “Incoherent meta-imaging system for noise-robust object recognition”, in [2024 conference on lasers and electro-optics pacific rim \(CLEO-PR\)](#) (2024), Mo1A4.
- ^{C2}D. Lee, J. Kim, H. Park, et al., “Design of correlated photonic time disorder for unidirectional scattering”, in [Advanced photonics congress 2023](#) (2023), NoTu3C.4.
- ^{C3}J. Kim, S. Park, D. Lee, S. Yu, and N. Park, “Data-driven engineering of active photonic disorder”, in [Frontiers in optics + laser science 2022 \(FIO+LS\)](#) (2022), JW4A.20.
- ^{C4}D. Lee, J. Kim, N. Park, and S. Yu, “Molecular dynamics for microscopic analysis of refractive index in amorphous hafnium oxides”, in [Frontiers in optics + laser science 2022 \(FIO+LS\)](#) (2022), FW7C.5.

^{C5}J. Kim, S. Yu, and N. Park, “Neural-network-based design of tunable multilayer films”, in [OSA advanced photonics congress 2021](#) (2021), JW4B.3.

HONORS AND AWARDS

Feb 2023 Distinguished Dissertation Award, ECE, SNU
Feb 2019 Best Poster Award, OSK Winter Meeting
2014–2016 National Science & Engineering Undergraduate Scholarship, Korea Student Aid Foundation

ACADEMIC SERVICES

Reviewer *Optica, Photonics Research, Optics Express, Optics Letters*
 Nature Electronics, Nature Communications
 ACS Photonics, Advanced Functional Materials